How heavily are PhD students offloading their scientific writing to AI? Evidence from US STEM PhD dissertations*

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Abstract:

Large language models are diffusing rapidly into science, raising concerns that delegating research writing to AI may undermine the development of expertise that scientific training is intended to build. We measure AI-generated writing in the dissertations of a near-census of US STEM PhD graduates from 2019 to 2026 and link graduates to their subsequent employment. AI writing is absent before 2023, present in 29% of dissertations filed in 2026, and rapidly growing. Use is more common among students from non-English-speaking countries and at lower-ranked programs. Within the same program and cohort, graduates whose dissertations contain AI writing are less likely to enter academic research careers and more likely to enter industry, with this pattern strongest among US-origin students. Even as AI may raise research productivity, it thus appears to also be displacing part of the work through which scientific expertise is traditionally developed. Despite this, few universities have AI policies we can identify that address dissertation research explicitly, and fewer than 5% of 2026 dissertations disclose AI use.

Keywords: Artificial intelligence, Doctoral training, Scientific workforce, Higher education, Knowledge collapse

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Introduction

Artificial intelligence (AI) technologies are rapidly diffusing across science and society (Chatterji et al. 2025, Handa et al. 2025) and increasingly reshaping how science is done (Gao and Wang 2024, Agrawal et al. 2026, Hao et al. 2026, Codreanu et al. 2026). Although AI takes many forms that are being used for different scientific ends—from predicting protein structures (Jumper et al. 2021, Varadi et al. 2022) to resolving mathematical conjectures (Alon et al. 2026)—an increasingly common manifestation of AI in science is the use of large language models (LLMs), as a growing share of research papers are conceived, executed, and partially or fully written by LLMs (e.g., Boiko et al. 2023, Liang et al. 2025, Kobak et al. 2025, Kusumegi et al. 2025, Walther and Dutordoir 2025, Holzwarth et al. 2026, Lu et al. 2026, Siler 2026, Gartenberg et al. 2026).

Enthusiasm about the potential for AI to accelerate the pace of discovery is substantial (e.g., Cockburn et al. 2019, Bianchini et al. 2022, Wang et al. 2023, Amodei 2024, Gil and Moler 2025, Dolgin 2026). However, as LLM use in science spreads, concerns are mounting over the pressures it puts on the scientific enterprise. Much of this concern centers around research evaluation and quality control vis-à-vis peer review (e.g., Liang et al. 2024, Naddaf 2025, Gibney 2026, Thorp 2026a,b, Nature 2026). Other observers have questioned the future of the research paper as a scientific unit (e.g., Mollick 2024, Cowen 2026a,b, Nature 2026), and some have even begun to reconsider who or what research is for if far more knowledge is being produced than can be consumed (Jones 2009, Chu and Evans 2021, Messeri and Crockett 2024, Liu et al. 2026).

A more fundamental threat, however, may be to scientific expertise and human-embodied knowledge (Acemoglu et al. 2026). This is particularly the case for current and future doctoral students, who represent the next generation of scientists. Much of an individual scientist's research ability is tacit and learned through experience (Ma et al. 2020), with the core of this knowledge cultivated in graduate school, where students have traditionally developed skills and taste through slow, arduous, hands-on work pitted with failures and continual iteration. Like research technologies of the past (e.g., computers), AI makes it possible to operate at a higher level of abstraction than previously possible, potentially cutting out much of this work and putting at risk the expertise this experience generates (Bastani et al. 2025, Strömberg et al. 2026, Autor et al. 2026). LLMs' cost and speed make them an especially attractive substitute for this effort. Intensifying competition at conferences and journals—itself a symptom of AI—merely reinforces these incentives, creating a feedback loop that only further encourages using AI to stay competitive on output and other evaluative metrics on which academic scientists' careers are built.

This paper examines how heavily AI is being used by PhD students, who is using it, and how it

relates to students' professional outcomes—a potential proxy for new scientists' quality of training and depth of expertise. Following recent literature, we focus on AI-generated writing, evaluating the AI content of PhD dissertations in a nearly-complete sample of US STEM PhD graduates since 2019, and comparing career trajectories of students who were more versus less reliant on AI in their dissertation work. We argue that AI-generated writing is a particularly important endpoint for evaluating training effort, as the writing stage of research is where ideas must be stated precisely and in turn where failures in logic and method are often discovered and worked out. The experience of consolidating and communicating research may thus be one way in which students become more effective critical evaluators and improve their understanding of their own work. Delegating this effort to AI risks increasing short-term productivity at the expense of early career scientists' skill acquisition, scientific capabilities, and long-run research capacity.

Methods and Data

To evaluate these questions, we collected data on US-trained PhD graduates since 2019, applied AI detection tools to dissertation abstracts to assess AI use, and linked individuals to their LinkedIn profiles to evaluate post-PhD outcomes. While a small number of prior articles have assessed LLM writing in dissertation research in small samples—typically specific universities or fields, often assessed through surveys and interviews (e.g., Hoomanfard and Shamsi 2025, Akbar 2025, Sarofian-Butin 2026)—we perform this analysis at a national scale.

Our sample consists of STEM PhD graduates from US universities whose dissertations were filed in ProQuest Dissertations & Theses Global (PQDT) between 2019 and May 2026, which we extend by collecting data from institution repositories for universities not represented in PQDT over this period. The ≈226,000 graduates in this combined sample align closely with totals in the National Science Foundation's Survey of Earned Doctorates (an annual census of US PhD graduates) by year, institution, and field, with approximately complete coverage of US PhD graduates in the life, physical, and mathematical sciences and 85% coverage of engineering. We evaluate the AI content of dissertations using Pangram 3.3.2 (Emi and Spero 2024), a validated commercial AI detection service which is increasingly used in research for AI detection (see, e.g., Jabarian and Imas 2025 for validation, and Gartenberg et al. 2026 for an application). We process all abstracts and a stratified random sample of complete dissertations (stratifying over fields and years), which we use to verify that AI-generated abstracts are a surrogate for AI-generated writing in the full text—consistent with prior research which has found the same (e.g., Gartenberg et al. 2026). We additionally process the full text of nearly all dissertations to identify and extract acknowledgments of AI use as indicators of AI disclosure (accessed using ProQuest's TDM Studio application, through which dissertations' full text is available for 92% of our PQDT sample; see appendix for details).

We identify graduates' post-PhD employment outcomes by linking them to LinkedIn profiles (via Revelio Labs data), leveraging a linking infrastructure developed in prior work (Shvadron et al. 2025). We also measure high school and baccalaureate country for a subset of graduates, which we use as a proxy for likely country of origin and to identify potential non-native English-speakers. Using metadata on the graduates in our sample from PQDT and repositories, we also identify degree characteristics such as institution and degree subjects, which we group up to major fields. We use data from the Integrated Postsecondary Education Data System (IPEDS) to measure institution characteristics and from National Research Council (2011) to measure individual program rankings, which we merge into our graduate sample on institution and field.

We supplement these data by collecting information on AI policies that may apply to dissertation research at the 146 R1 institutions in our sample (based on the 2021 Carnegie Classification). We gather university and graduate school policy documents from each institution as of October 2026 and categorize policies by their context (e.g., policies for dissertations specifically, research generally, coursework, data security, etc.) and vis-à-vis specific requirements.

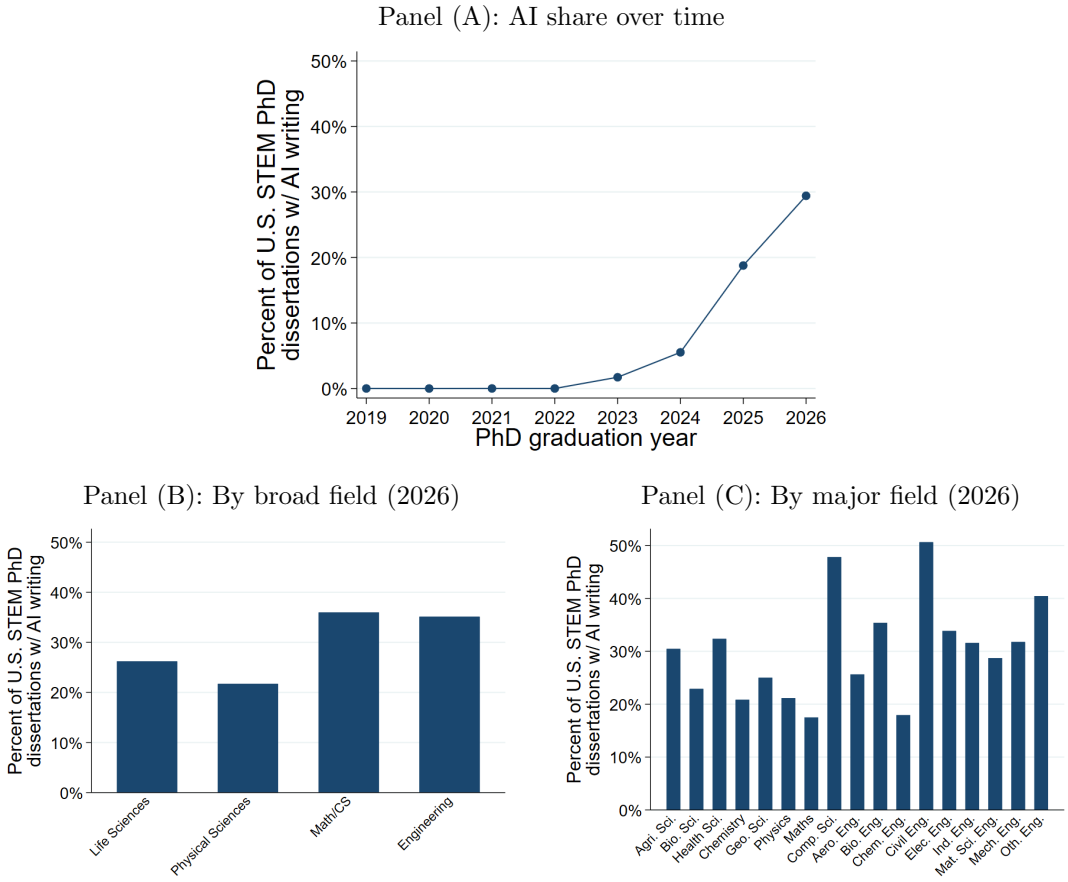
Nearly 30% of dissertations have AI-generated content

Figure 1 reports the share of US STEM dissertations with AI-generated abstracts, by year of degree (Panel A) and by field (Panels B and C, for 2026 graduates only). Through 2022, the AI share is effectively zero: across roughly 126,000 dissertations filed between 2019 and 2022, Pangram flags a total of six abstracts as AI-generated. Because these cohorts wrote dissertations before LLMs were publicly available, these pre-2023 values serve as a validation test of the detection method and indicate a false positive rate of less than 0.01%. AI writing first appears among 2023 graduates (1.7% of dissertations), then grows to 5.5% in 2024, 18.8% in 2025, and 29.4% in the 2026 cohort, where our data include dissertations filed through May 2026.

Benchmarked against recent research on AI use in scientific writing (which primarily examines peer-reviewed journal articles and pre-prints), PhD research appears to be lagging wider scientific activity in AI usage by 1-2 years: Liang et al. (2025) and Kobak et al. (2025) previously estimated that 13-22% of 2024 abstracts in computer science and biomedicine (respectively) were LLM-modified, whereas our corresponding estimate for 2024 dissertations is ≈5%. Reweighting our measures of AI use in abstracts to predict use in full text (using conditional probabilities of Pr(AI in full text | AI in abstract) in our full text subsample) yields only modest changes: we project that ≈8% of dissertations were likely to have AI writing in 2024 and 31% in 2026.

Panels B and C show that adoption is uneven across fields. Evaluated across broad fields (per the Classification of Instructional Programs (CIP) taxonomy), in the 2026 cohort, AI-generated

Figure 1: How heavily is AI being used in dissertation research?



Notes: Panel (A) shows the annual share of dissertations whose abstracts were AI-generated, according to Pangram detection. The evaluated AI share was 0 through 2022 but has been rapidly accelerating since then, and is nearly 30% in our 2026 sample (covering dissertations filed with ProQuest through May 2026). Panels (B) and (C) show the share of dissertations in our 2026 sample with AI abstracts, by broad field and major field.

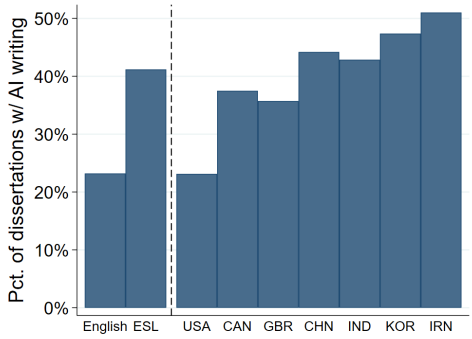
abstracts are most common in the mathematical and computer sciences (36%) and engineering (35%), followed by the life sciences (26%) and physical sciences (22%). These averages mask a wider spread across more specific fields, from 17% in mathematics and statistics and 18% in chemical engineering to 48% in computer science and 51% in civil engineering, with other engineering disciplines, health sciences, and agricultural sciences in between (29-40%). The fields with the most intensive AI use in dissertation research are those in which LLM adoption in published research has also been fastest (Liang et al. 2025, Kousha and Thelwall 2026); those at the bottom (e.g., chemistry and physics) are fields whose abstracts are more likely to include specific notation, compounds, and quantities. The same rank-order holds among 2025 graduates (engineering 24%, math/CS 24%, life sciences 16%, physical sciences 14%), suggesting that this heterogeneity reflects field-specific differences in the pace of AI adoption rather than its likely eventual reach.

Dissertation research remains human-generated at the top

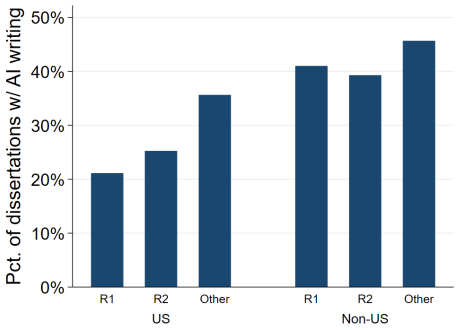
Use of AI in dissertation research writing varies across students and institutions. Figure 2 further disaggregates AI use by likely country of origin and native language, the research intensity of their PhD institution, and the National Research Council rank of their doctoral program. We proxy for origin country with the country of a graduate's high school or baccalaureate degree, observed in linked LinkedIn education histories for roughly 60% of the sample.

Figure 2: Which PhD students are using AI to write their research?

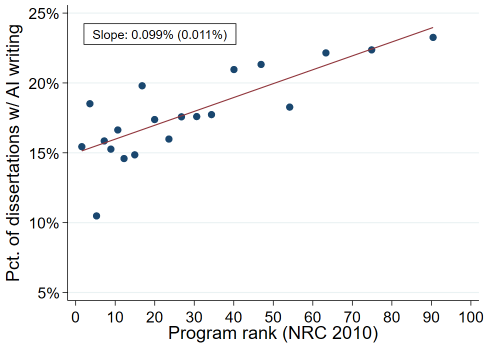
Panel (A): AI use by language and country



Panel (B): AI use by Carnegie Class



Panel (C): AI use by program rank



Notes: Panel (A) shows graduates' tendency to use AI to draft dissertation research by whether they are from an English-speaking country, and by country for select high-volume countries. Panel (B) shows this tendency by institution research intensity (measured by its 2021 Carnegie Classification), separately for US and foreign origin students. Panel (C) shows this tendency by program rank (measured using National Research Council 2010 rankings; see appendix for details). Sample in Panels (A) and (B) consists of 2026 dissertations available through May 2026; sample in Panel (C) consists of 2025-2026 dissertations. English as a second language students and students at lower research intensity institutions and lower-ranked programs are significantly more likely to substitute AI writing.

Non-native English-speaking students use AI far more intensively than native speakers.[1] Among

[1]Note that although early research on use of LLMs (especially ChatGPT) in scientific writing suggested that AI classifiers are more likely to flag non-native, human-generated English prose as machine-generated (Liang et al. 2023), this does not appear to be a concern for modern detectors (like Pangram) or in our context. Among 2019-2022 graduates (pre-LLMs), abstracts by students from non-English-speaking countries are flagged as AI-generated at a 0% rate (identical to those by English-speaking peers), on samples of roughly 18,000 graduates per year. The gap we observe emerges only after 2022 and grows with adoption.

2026 graduates, 41% of dissertations by students educated in non-English-speaking countries have AI-generated abstracts, against 23% for students from English-speaking countries (Panel A). The gap is present for every major sending country: 44% for students from China, 43% from India, 49% from South Korea, and 51% from Iran, versus 23% for those with US lower degrees—a benchmark which is already potentially biased upwards by the inclusion of foreign-born students with US baccalaureates. The gap has also widened each year as adoption spreads: non-English-speaking students' rate of AI writing was 2.8% in 2023 against 0.9% for English-speaking students, 8.3% against 3.7% in 2024, and 28% against 14% in 2025. In regressions with institution-field fixed effects, which effectively compare students in the same doctoral program, the difference is 1.5, 3.7, and 11.7 percentage points (p.p.) in the three periods (respectively), and the gap remains largest for students from Iran (23 p.p. in 2025-26), South Korea (15 p.p.), China (13 p.p.), and India (11 p.p.), whereas students from Canada and the UK do not differ from US-origin students in AI use. These patterns are consistent with evidence from studies on published research, where LLM-assisted writing is most prevalent among authors in non-English-speaking countries (Kobak et al. 2025, Liu et al. 2025, Kusumegi et al. 2025), and it is consistent with the presumption of LLMs being most valuable to students for whom English writing is most costly (Amano et al. 2023).

AI use also increases at institutions with lower research intensity and programs of lower rank. In the 2026 cohort, 28% of dissertations at R1 institutions have AI writing, compared with 31% at R2 institutions and 37% at other doctoral institutions, and this gradient is particularly strong among US-origin students (foreign-origin students write with AI at high rates at all institutions; Panel B). Within R1 institutions, the relationship with program quality is roughly monotone: among 2025-26 graduates of programs ranked in the top ten of their field, 15% have AI-generated abstracts, rising to 22% for programs ranked 40-50 and 26% for programs ranked 90-100, a gradient of about one p.p. per ten rank positions (Panel C). Roughly a third of this gradient reflects the composition of top programs, which enroll more international students; conditioning on origin, the slope falls from 0.10 to 0.07 p.p. per rank position but remains precisely estimated.

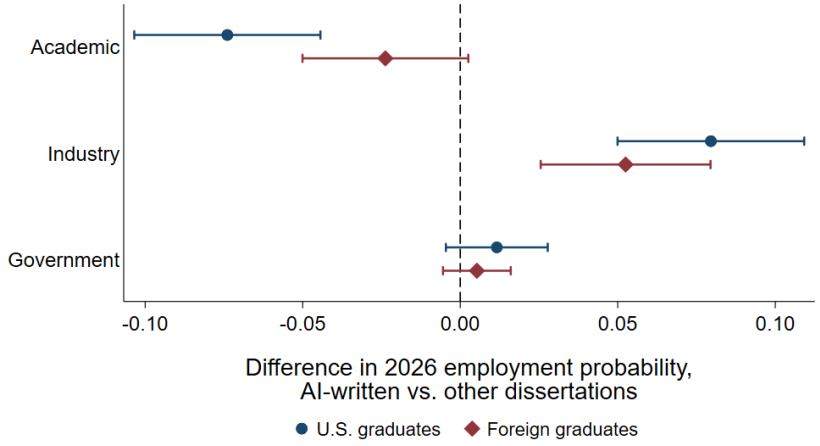
These differences are stark at the extremes. Among R1 institutions with at least 50 graduates since 2025, the AI share is 9% at MIT, 10% at Caltech, and 13% at Princeton, versus 30%+ at the ten institutions with the highest rates (Appendix Table B.1). Several top-ranked programs—e.g., mathematics at Berkeley, bioengineering at Northwestern, chemistry at Ohio State—have no AI-generated abstracts at all, while the highest institution-field cells exceed 50% (Appendix Table B.2). AI use also clusters within programs and advisors: a 10 p.p. higher rate of AI writing among the previous year's graduates of the same doctoral program is associated with a 3 p.p. higher probability that a student's own writing is AI-generated—patterns which may reflect peer effects, common

access, shared norms, or other shared features. Similar correlations hold among students with the same primary advisor or committee chair (Appendix Table B.3).

Graduates with AI-written dissertations tend to enter industry

Assessing training quality, and how it varies with AI, remains difficult. One potential surrogate endpoint is a graduate's post-PhD employment outcome. Using data on graduates' LinkedIn employment histories, we evaluate differences in the current (2026) employment of 2023-2025 PhD graduates of R1 institutions, allowing at least half a year after graduation for LinkedIn profiles to update, and compare graduates with AI-written dissertations to others of the same institution and field. Because AI use is concentrated among foreign-origin students, who on average also differ in their post-PhD employment opportunities, we estimate the relationship separately for US- and foreign-origin graduates.[2] The results are presented in Figure 3.

Figure 3: Differences in employment patterns among graduates with AI-written dissertations



Notes: Figure shows the differential tendency for graduates with AI-written dissertations to be employed in academia, industry, or government in 2026. Differences estimated over graduates who obtained their PhD at R1 institutions between 2023 and 2025 (allowing time for LinkedIn profiles to update post-graduation). Estimates obtained from a regression of the employment outcome on indicators for (i) AI writing, (ii) foreign lower degree (a proxy for non-US origin), and (iii) their interaction. All regressions include institution-field and year fixed effects, with standard errors clustered by institution-field; results are similar in magnitude and precision with institution-field-year (i.e., program cohort) fixed effects. Error bars represent 95% confidence intervals.

Among US-origin graduates, those with AI-written dissertations are 7.4 p.p. less likely to be employed in academia in 2026 (17% over the 44% base rate) and 8.0 p.p. more likely to be employed

[2]Specifically, we regress each employment outcome on an indicator for an AI-generated abstract, an indicator for a foreign lower degree, and their interaction, with institution-field and graduation year fixed effects and standard errors clustered by institution-field. The figure plots the marginal effect of AI writing for each origin group. Sectors are not mutually exclusive: a graduate who changed jobs during 2026 can appear in two, and "academic" employment includes universities, teaching hospitals, and federally funded research centers.

in industry (16% over a 50% base), with no difference in government employment. These differences attenuate substantially for likely foreign-origin graduates. We also find differences within academia: conditional on academic employment, graduates with AI-written dissertations are 5.4 p.p. more likely to hold non-tenure-track positions (on a base of 12%), with no significant difference in the likelihood of tenure-track or postdoctoral positions. AI writing thus appears disproportionately common among graduates who exit an academic research career.

Importantly, this analysis is correlational (not causal) and supports multiple interpretations. Interpreted causally, it suggests that graduates who write with AI may have a training deficit that leaves them unprepared for a research career. On the other hand, a plausible interpretation is reverse causal: late-stage students who do not seek a research career may write lower-effort dissertations. If so, we might expect other markers of low effort, including shorter or more poorly written dissertations. Empirically, however, dissertation length (measured in pages) does not correlate with AI use, and employment differences in Figure 3 persist controlling for length. On balance, we view the evidence as suggesting an adverse impact of AI use on research capability.

There are two reasons why the differences in Figure 3 may be conservative. First, our abstract-based AI detection captures only one part of an AI-written dissertation (Appendix Tables A.1 and A.2), and this measurement error may bias magnitudes downwards. The true relationship between full-text AI use and employment outcomes may be materially larger. Second, the sample is restricted to R1 graduates, who comprise a large majority of our sample but among whom both (i) research careers are more common and (ii) AI use is comparatively lower.

Policies and norms governing AI use are still emerging

Whether students are using AI with their institutions' explicit or implicit sanction remains an open question. Figure 4 summarizes policies we can identify at R1 institutions as of October 2026 and the frequency with which dissertations acknowledge AI use. All 146 R1 institutions in our sample have some central rule that relates to AI use by students,[3] but only about one quarter (38) have a policy for dissertation research specifically (Panel A). Of the dissertation policies we can identify, none prohibits AI use outright, and requirements of approval (by the student's advisor or committee) and disclosure are common (Panel B). Despite this, disclosure is in practice rare: among 2026 dissertations, fewer than 5% acknowledge AI use (Panel C).[4] Juxtaposed against the

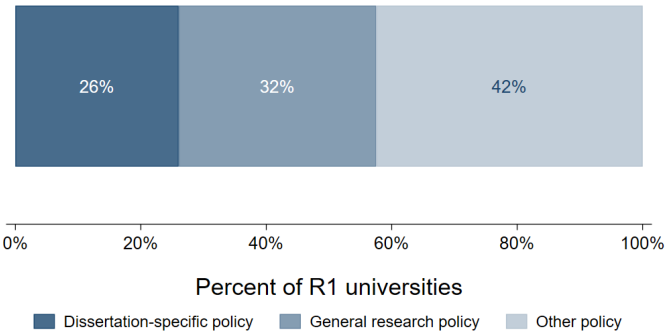
[3] In decreasing order of specificity to dissertation research: 84 institutions have explicit policies governing AI use in dissertations or other research, 25 have an honor code that subsumes this question, and 26 have rules for coursework broadly but not for research explicitly. For the remaining 11 (7.5%) of these institutions, the most specific provisions govern data security or leave policies over AI use to instructors' discretion.

[4] One-quarter of those come from just one institution (North Carolina State University), whose dissertation template includes an authorship statement page with a placeholder for reporting AI use.

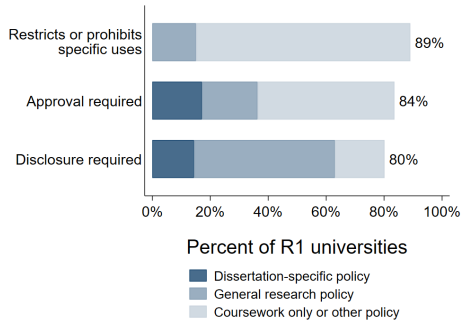
earlier evidence that roughly 30% of graduates in 2026 used AI to write at least a portion of their dissertation, the gap suggests that policies, norms, and enforcement mechanisms are substantially lagging the changing reality of modern scientific training.

Figure 4: AI policies across R1 institutions and AI disclosure patterns

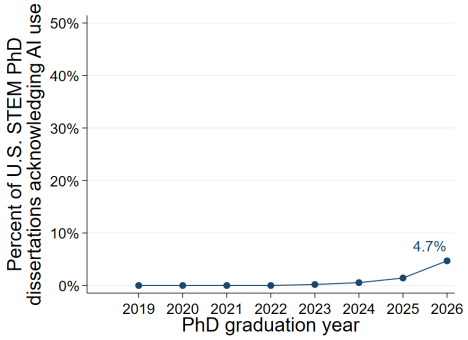
Panel (A): How is AI use in dissertation research governed? (R1 only)



Panel (B): Specific requirements (R1 only)



Panel (C): AI disclosure rates (full sample)



Notes: Panel (A) shows the share of R1 institutions in our sample with AI policies of various types as of October 2026; the "Other policy" category subsumes policies covering coursework generically, data security, or providing instructor discretion. Panel (B) shows the share of these institutions naming specific requirements such as restrictions on AI use, approval requirements, and disclosure requirements in doctoral research, other research, and other contexts. Note that most use restrictions come from policies governing data security rather than academic use. Panel (C) shows actual detected disclosure rates in our dissertation sample, by year, visualized on the same scale as Figure 1(A) for comparability.

Discussion

Taken together, these patterns suggest the US scientific training system is stratified in observable ways with respect to AI use: students at lower-tier institutions, and non-native English speakers, are disproportionately delegating their writing to AI. If research communication—especially the act of writing—is important to scientific training, then the growing use of AI we document suggests a growing disparity in the quality of PhD training across these categories.

A natural counterargument, however, is that the benefits of AI may flow in the other direction, narrowing performance gaps. Non-native English speakers face large, well-documented costs to

writing science in English (Amano et al. 2023), and the experimental literature finds that generative AI helps more disadvantaged writers most (Noy and Zhang 2023, Brynjolfsson et al. 2025), narrowing gaps in readability and citations for scientists at non-English institutions (Kaniel et al. 2025, Lin et al. 2025). It may also be the case that both patterns are true: AI may close gaps in writing quality or comprehension while also removing the impetus for the careful thinking which writing requires. The evidence in this paper is consistent with both, as PhD graduates with AI writing in their dissertations are less likely to remain in academic research, but this difference is attenuated for those ostensibly from non-native English-speaking countries.

Though the evidence in this paper is suggestive of a cost, the endpoints we study ultimately cannot fully resolve whether PhD students' use of AI in scientific writing is in fact degrading their preparation for a research career. Observers who see the act of writing as forcing a precise understanding are likely to believe so, while those who view scientific communication as an instrumentality fully separable from research itself may disagree. Empirically, it is likely too soon to know for sure: indications of degraded research capability—such as evidence that these students are less likely to publish independently, or win competitive research grants—will take more time to accumulate, and causal evaluative designs will also be needed to close the question.

More generally, we are still in the early days of AI, and much may yet change. Adoption is clearly growing, and at its current growth rate, a majority of US STEM dissertations will have AI-generated content by 2028. AI technology is also rapidly improving, and institutions are evolving with it, albeit more slowly: as Figure 4 shows, few institutions appear to have dissertation AI policies in place thus far. What is clear to us, however, is that even as AI holds the potential to unlock new discoveries and dramatically accelerate progress, it is changing the returns to human capital investment and challenging the case for developing deep scientific expertise through slow, careful scholarship (Gartenberg et al. 2026). Economic theory suggests sustaining students' incentive to do the hard work themselves will require either changes in how scientists are evaluated and what is professionally rewarded, or verification technologies to measure and enforce human-generated effort—or both. Unlike peer-reviewed journals, however, doctoral degrees already have a verification institution in the form of a live defense, whose importance as a test of expertise and understanding—and as a condition for degree conferral—may grow. Ultimately, whether the next generation of scientists is trained or merely credentialed remains to be seen. The answer will reveal itself over time, but research in the interim may offer leading indications.

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Online Appendix

How heavily are PhD students offloading their scientific writing to AI? Evidence from US STEM PhD dissertations

Daniel P. Gross, Dror Shvadron, and Hansen Zhang

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A Data Appendix

This appendix describes our data sources, measurement, and analytical methods. Note that portions of this appendix—particularly those documenting the ProQuest sample and our approach to linking graduates to their post-PhD employment—overlap with the supplementary material to Shvadron et al. (2025a) and Shvadron et al. (2025b), where we use a partially overlapping sample and partially overlapping measures to study different questions.

A.1 Base sample: ProQuest dissertations + university repositories

The sample for this paper consists of U.S. STEM (Science, Technology, Engineering, and Mathematics) PhD graduates who received their degrees between January 2019 and May 2026 (the time of data collection), providing us roughly four graduate cohorts prior to the public release of ChatGPT (in November 2022) and four post-ChatGPT cohorts.

Our core data source for this sample is ProQuest Dissertations & Theses Global (PQDT), a commercial database of higher degree graduates and their theses, from which we obtain graduate metadata.[1] At the time of acquisition, the PQDT database contained over six million global master's theses and doctoral dissertations, including in the natural sciences, engineering, humanities, social sciences, and other domains. PQDT data report several characteristics of the graduates in its sample, including their name, degree year, degree name, university, dissertation characteristics (title, abstract, and subjects), advisor(s), and more. Following our own prior work, we filter the PQDT corpus to U.S. STEM PhD graduates as follows. We first removed (i) non-U.S. dissertations and (ii) non-doctoral theses and dissertations. We also manually reviewed degree names to identify and restrict to those corresponding to research degrees in the natural sciences and engineering.[2] We then crosswalk these graduates' self-reported subjects (as provided in the PQDT data) to "major fields" defined by the U.S. National Center for Science and Engineering Statistics (NCSES) and filter our sample to dissertations in 17 major fields which are typically considered STEM subjects.[3] We extend the PQDT sample by retrieving analogous data from digital thesis repositories of universities which are not represented in PQDT in our target years, for as many such universities as we could crawl online, compiling a similar record (name, degree year and name, dissertation characteristics, advisors) and filtering and mapping subjects in analogous ways.

[1] ProQuest is the leading distributor of U.S. dissertations, which it re-publishes under license, and its data have been used in numerous prior studies of scientific trainees (e.g., Toole and Czarnitzki 2010, Bikard et al. 2015, Buffington et al. 2016, Jiang et al. 2023, Antman et al. 2023, Arora et al. 2026, among others).

[2] Though most doctoral degrees are PhDs, PQDT also includes other doctorates such as DScs or field-specific doctorates (e.g., D.CS. (Doctor of Computer Science) or Chem. D. (Doctor of Chemistry)), as well as certain doctoral professional degrees (e.g., PsyD (Doctor of Psychology) or D.N.P. (Doctor of Nursing Practice)).

[3] Our focal major fields are: Agricultural sciences and natural resources, Biological and biomedical sciences, and Health sciences (belonging to the broad field "Life Sciences"); Physics and astronomy, Geosciences, atmospheric sciences, and ocean sciences, and Chemistry (belonging to "Physical Sciences"); Computer and information sciences and Mathematics and statistics (belonging to "Mathematical and Computer Sciences"); and Aerospace, aeronautical, and astronautical engineering, Bioengineering and biomedical engineering, Chemical engineering, Civil engineering, Electrical and electronics engineering, Industrial and manufacturing engineering, Materials science engineering, Mechanical engineering, and Other engineering (belonging to "Engineering").

A.1.1 Additional person characteristics

Following the linking procedures described in the next section, we link graduates in this sample to LinkedIn profiles, using data from Revelio Labs and matching on name and PhD degree details. We use these links to obtain individuals' complete education histories, identify reported high school and baccalaureate degrees, and retrieve the country of each degree. We in turn use these lower degree countries to measure whether a graduate is likely to be from a foreign (non-US) country, and to categorize those countries into native English speaking countries (USA, Canada, UK, Australia, New Zealand, Ireland) versus others. Though imperfect, this approach is conservative, prioritizing precision over recall: in a U.S. PhD student sample, non-US high school and baccalaureate degrees are overwhelmingly obtained in students' native countries (such that when observed, we can precisely and confidently infer the origin country), but we may fail to measure as foreign those PhD students who completed their baccalaureate degree in the U.S. (reducing recall). This mismeasurement will tend to attenuate English/non-English and US/non-US comparisons, which implies any such differences are likely a lower bound.

A.1.2 Additional program characteristics

We manually crosswalk degree institutions to Integrated Postsecondary Education Data System (IPEDS) UnitIDs, the institutional identifiers which SED, IPEDS, and other U.S. higher education datasets are identified on. We use IPEDS to retrieve institutions' 2021 Carnegie Classification, grouping them into R1 ("Doctoral Universities: Very High Research Activity"), R2 ("Doctoral Universities: High Research Activity"), and others (a catch-all which consolidates professional doctoral universities, PhD-granting research institutes, and any others).

We supplement these data with measures of university and program rank. We obtain general university rankings from U.S. News & World Report (where available; excludes specialty institutions) and doctoral program rankings from the National Research Council (2011) publication "A Data-Based Assessment of Research-Doctorate Programs in the United States" and accompanying rankings tables.[4] This project produced rankings of over 5,000 doctoral programs in 62 fields at 212 U.S. universities, based on a 2005-2006 program characteristics. The data include both survey-based rankings and statistical rankings with field-specific estimated weights on different program characteristics, with each ranking bootstrapped and the accompanying files reporting the bottom 5% and top 95% rank of each program across bootstrap iterations. We calculate the average regression-based rank for each program (averaging between its 5th and 95th percentile bootstrap ranks) and use this as our measure of program quality. To link the NRC data to our sample, we crosswalk universities to IPEDS, NRC fields to our canonical major fields, and link on university-field; where a university has multiple programs in the same major field, we measure its rank as the best rank reported among any of the associated programs.

[4]See https://www.nationalacademies.org/projects/BHEW-Q-03-02-A/publication/12994 for the report and data.

A.1.3 Completeness/representativeness of sample

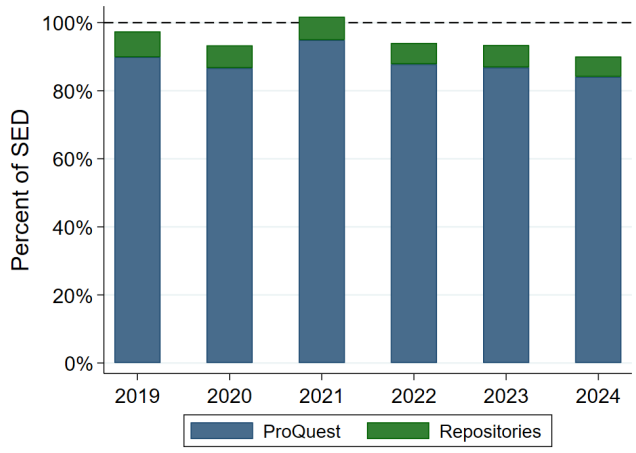
To evaluate the completeness and representativeness of our data, we compare counts of PQDT + Repository (henceforth, "PQDT+") graduates to counts of U.S. doctoral graduates from the Survey of Earned Doctorates (SED), an annual census of research doctorate recipients from U.S. universities which has been administered by NSF since 1958, with >90% completion rates. Although modest differences in counts may arise due to differences in field assignment—SED similarly assigns graduates to fields based on self-reported subjects, but we do not have access to their mapping technique—it provides an administrative benchmark for our sample.

We make several comparisons between PQDT+ and SED, including on annual counts, totals by broad field for our study period, and correlations at different units of aggregation. Figure A.1 shows that our PQDT+ sample represents 90-100% of SED STEM graduates in every year for which we have data (the most recently-reported SED year is 2024), with PQDT comprising 85-95% of the SED population and repository data filling in another ≈5%. That the PQDT+ sample slightly exceeds SED in one year (2021) can be attributed to one of three possibilities: that (i) PQDT+ was more complete than SED this year (as SED response rates are very high but less than 100%), (ii) differences in timing (PQDT+ reports when dissertations were made available, SED when students graduate), or (iii) modest deviations in how we map subjects to fields versus how SED performs this mapping which can lead to modest sample differences.

Figure A.2 evaluates completeness by broad field, and indicates that the PQDT+ sample returns approximately complete sampling of life science, physical science, and mathematical/computer science graduates, and very high (85%) sampling of engineering graduates; that Math/CS slightly exceeds 100% is potentially attributable to the same reasons, as well as fungibility in the coding of some subjects to computer science versus electrical engineering.

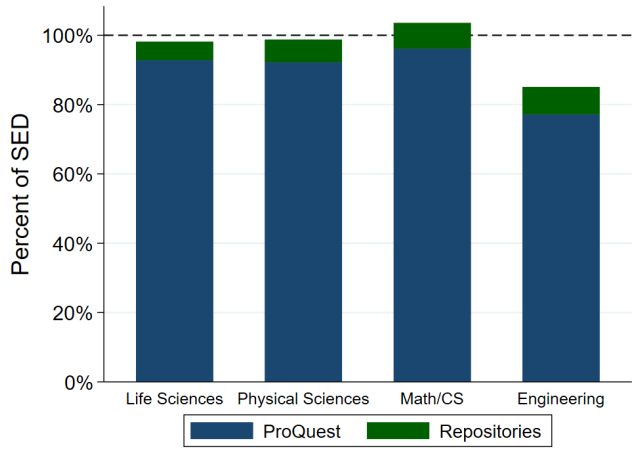
Figure A.3 provides binned scatterplots of SED vs. PQDT+ graduates at the university level, university-year level, field-year level, and university-field-year level (Panels A to D, respectively), including bivariate regression coefficients. At all levels of aggregation, PQDT+ and SED counts closely track each other and cluster around the 45-degree line.

Figure A.1: Completeness of sample, by year (benchmark: SED)



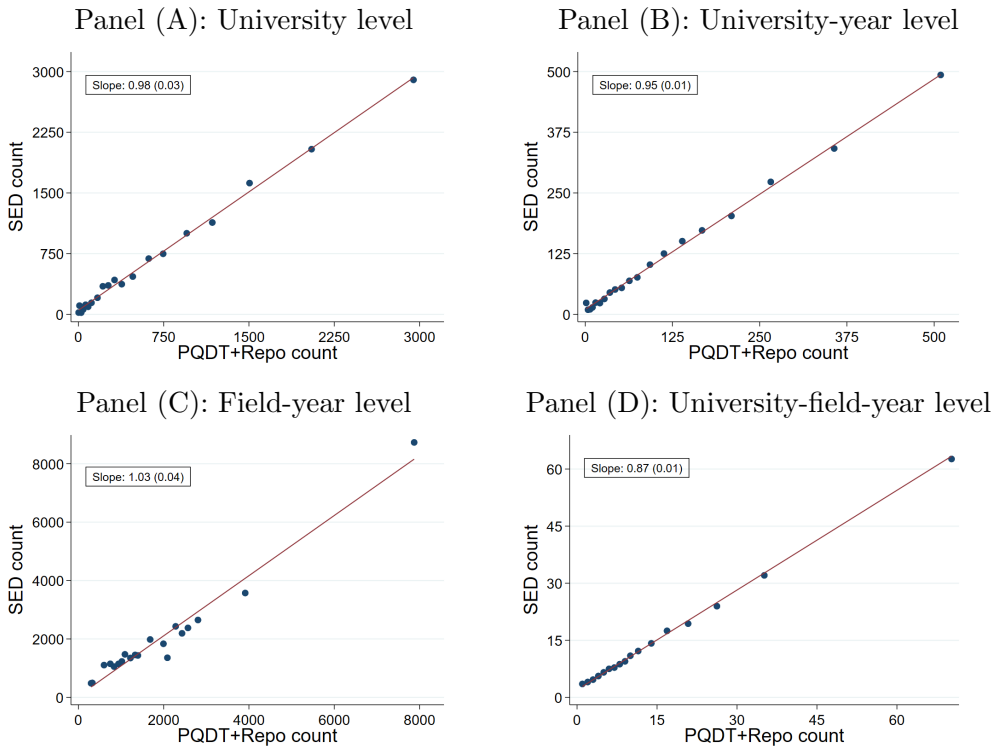
Notes: Figure evaluates the completeness of our sample across years, using the SED as a benchmark. Each bar reports the number of U.S. STEM PhD graduates in our PQDT sample (blue) and university repository backfill (green) relative to the SED total. Graduates in each of these data sources (PQDT, repositories, SED) are associated to fields as described in text. SED data currently available through 2024.

Figure A.2: Completeness of sample, by broad field (benchmark: SED)



Notes: Figure evaluates the completeness of our sample across broad fields, using the SED as a benchmark. Each bar reports the number of U.S. STEM PhD graduates in our PQDT sample (blue) and university repository backfill (green) relative to the SED total. Graduates in each of these data sources (PQDT, repositories, SED) are associated to fields as described in text. SED data currently available through 2024.

Figure A.3: Correlation of PQDT+ vs. SED graduate counts at varying units of analysis



Notes: Figure shows a binned scatterplot of PQDT + Repository vs. SED graduate counts in our sample at each of four levels: university (Panel A), university-year (Panel B), field-year (Panel C), university-field-year (Panel D). Graduates are associated to fields as described in text. Sample covers STEM graduates of universities in the continental U.S. between 2019 and 2024 (the last year of available SED data).

A.1.4 Measuring dissertation AI content

We retrieve dissertation abstracts for all of the 226,911 graduates in our sample from PQDT and university repositories, and full dissertation text for a subsample of PQDT-listed graduates, randomly sampling 5 dissertations per major field (17 fields) per year between 2023 and 2026 (4 years). We then use Pangram (validated commercial AI detection service which is used frequently in similar research) to assess the AI content of these inputs, applying the Pangram v3.3.2 model through its bulk API, with discounted research pricing. Around 300 dissertations did not have abstracts or could not be scored by Pangram, leaving us with 226,591 abstracts and 340 randomly sampled full text dissertations that we assess for AI writing.

Table A.1 uses the full text sample to validate the abstract-level measures. We estimate a simple bivariate regression of full text AI use against abstract AI use:

$$\text{Full text AI}_i = \beta \cdot \text{Abstract AI}_i + \varepsilon_i \tag{A.1}$$

We evaluate this relationship at the individual level (Columns 1 to 3), then at more aggregated levels: university-year (Columns 4 to 5) and field-year (Columns 6 to 7). The table shows that a

dissertation with an AI-generated abstract is 55 percentage points (p.p.) more likely to have AI detected in the full text (quadrupling the base rate), has 11% more AI content (nearly quintupling the base rate), and is 20 p.p. more likely to have at least 20% of its text be AI-written (6x the base rate). Similar patterns emerge when evaluated at higher levels of aggregation.

In Table A.2 we switch the left- and right-hand side variables, such that the parameters indicate the likelihood of an AI abstract given a certain amount of AI content in the full text (conveying the same correlation as Table A.1 in different form). Accordingly, the same patterns emerge but now can be interpreted as follows. A dissertation with any AI in its text is 56% more likely to have an AI-generated abstract; a dissertation with 10 p.p. more of its content AI-generated is 25 p.p. more likely to have an AI-generated abstract, and a dissertation where >20% of the full text is AI written has over 90% probability of having an AI-generated abstract.

Table A.1: Abstract AI writing predicts full dissertation AI writing

	Individual dissertations			Univ-year dissertations		Field-year dissertations	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Has AI	Pct. AI	>20% AI	Pct. w/ AI	Mean AI	Pct. w/ AI	Mean AI
Has AI in abstract	0.556***	0.111***	0.204***				
	(0.068)	(0.021)	(0.056)				
Pct. w/ AI in abstract				0.513***	0.118***	0.709***	0.129***
				(0.089)	(0.031)	(0.085)	(0.026)
N	339	339	339	215	215	68	68
R^2	0.31	0.28	0.16	0.28	0.25	0.60	0.54
Y mean	0.153	0.023	0.035	0.156	0.025	0.153	0.023

Notes: Table shows that (i) AI writing in a dissertation's abstract significantly predicts whether it has AI writing in the full dissertation, and (ii) the AI-generated share of the abstract predicts the AI-generated share of the full dissertation. The sample for this analysis is a stratified random sample of 340 dissertations filed between 2023-2026, with 5 per major field and year, for which we retrieved the full text and evaluated its AI content using Pangram, as we did for all abstracts (note: one dissertation is associated with an institution we could not match to IPEDS and excluded from the estimation). The table presents analysis at multiple levels: at the individual level (Columns 1-3), university-year level (Columns 4-5), and field-year level (Columns 6-7). Outcomes are as follows: Column (1), whether a dissertation has AI in the full text; Column (2), the share of the full text that is AI-generated; Column (3), whether >20% of a dissertation's text is AI-generated; Columns (4) and (6), the share of dissertations in the cell that have AI in the full text; Columns (5) and (7), the average AI share of full text among dissertations in the cell. *, **, *** represent significance at the 0.1, 0.05, and 0.01 levels, respectively. Heteroskedasticity-robust SEs in parentheses.

Table A.2: Full text AI writing predicts abstract AI writing

	Individual abstracts			Univ-year abstracts		Field-year abstracts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Has AI	Has AI	Has AI	Pct. w/ AI	Pct. w/ AI	Pct. w/ AI	Pct. w/ AI
Has AI in full text	0.565***						
	(0.069)						
AI pct. of full text		2.514***					
		(0.339)					
>20% AI in full text			0.788***				
			(0.082)				
Pct. w/ AI in full text				0.552***		0.853***	
				(0.090)		(0.081)	
Mean AI pct. of full text					2.141***		4.155***
					(0.299)		(0.479)
N	339	339	339	215	215	68	68
R^2	0.31	0.28	0.16	0.28	0.25	0.60	0.54
Y mean	0.156	0.156	0.156	0.164	0.164	0.156	0.156

Notes: Table shows that (i) AI writing in a dissertation's full text significantly predicts whether it has AI writing in the abstract, and (ii) the AI-generated share of the full text predicts an AI-generated abstract. The sample for this analysis is a stratified random sample of 340 dissertations filed between 2023-2026, with 5 per major field and year, for which we retrieved the full text and evaluated its AI content using Pangram, as we did for all abstracts (note: one dissertation is associated with an institution we could not match to IPEDS and excluded from the estimation). The table presents analysis at multiple levels: at the individual level (Columns 1-3), university-year level (Columns 4-5), and field-year level (Columns 6-7). The outcome at the individual level is an indicator for whether an abstract is AI-composed; at the group level, the share of abstracts that are AI composed. *, **, *** represent significance at the 0.1, 0.05, and 0.01 levels, respectively. Heteroskedasticity-robust SEs in parentheses.

A.2 Measuring professional outcomes

We link graduates in our sample to career histories assembled by Revelio Labs from public professional profiles, principally LinkedIn, to observe their post-PhD employment. Revelio provides us what is effectively a mirror of the structured relational database behind LinkedIn profiles, periodically re-scraping the website to retrieve new users and profile updates. Revelio standardizes each profile in its database into a person record, a set of education records (institution, degree, field, and dates), and a set of position records (employer, title, start and end dates, and location), and resolves employers to a company identifier with associated metadata (web domain, industry code, parent company, and, for listed firms, securities identifiers). We access these data under license through Wharton Research Data Services, in the June 25, 2026 release.

A.2.1 Linking dissertations to career profiles

The dissertation data provide several pieces of fixed identifying information which can be used to link to professional profiles—chief among them are name and education history (degrees, institutions, and years). We use this information to make an initial set of candidate links between our sample and the Revelio data. This procedure begins with the manual construction of a crosswalk mapping PhD institutions in Revelio and PQDT to common identifiers (IPEDS UnitIDs), and then filtering Revelio profiles to those reporting a doctoral research degree (PhD or equivalent) at an institution also present in PQDT. Doing so yields 2,762,178 candidate profiles for the 1,375,136 dissertations in the PQDT base sample, prior to matching them on name.

We then split names in both sources into given, middle, and family names. Because raw name strings can have inconsistent order, punctuation, or inclusion of middle names and credentials, we perform this processing with batched LLM review followed by manual review and correction. A candidate pair is formed when the school matches, the degree year is consistent with the profile's education dates, and the names agree under one of seventeen matching rules. These range from exact agreement on all features, to hard-coded matching rules (agreement on initials, middle-name subsequences, and punctuated-corrected features such as a hyphenated given name split into two), to manually-constructed concordances of nicknames and for common romanization variants of Chinese family names (such as Lee/Li and Chang/Zhang), and finally to bounded fuzzy similarity. This process yields 642,699 candidate dissertation x LinkedIn profile pairs.

Every candidate pair receives an explicit accept or reject decision. Pairs matched under a deterministic rule are reviewed once; pairs matched under a fuzzy match or by an initials-only rule are evaluated through a succession of LLM reviews under prompts of varying strictness. We then subject the 620,798 pairs that survive these reviews to one further check for contextual consistency, which rejects 1,384 pairs (0.22%), leaving 619,414 accepted pairs.

Because pairs are reviewed independently, a dissertation can be accepted to multiple profiles and a profile to multiple dissertations. Of the accepted pairs, 546,229 are clean one-to-one assignments. The remaining profiles typically fall into the category of (i) dissertations linked to two profiles that appear to be duplicates of the same person (11,389 clusters), or (ii) dissertations linked to what appear to be two distinct users (15,221). We adjudicate these cases where possible, linking each dissertation to at most one profile, and dropping any candidate pairs which we are unable to disambiguate. The final match assigns 584,178 dissertations (42.5% of the PQDT base) to 581,318 unique Revelio users. Among the 2019-2026 graduates studied in this paper, 159,297 of 226,911 (70.2%) are linked to a Revelio profile. We use the linked profiles for two purposes. The first, described above, is to retrieve graduates' lower-degree institutions and countries as a proxy for country of origin. The second is to measure employment in 2026.

A.2.2 Employment in 2026

For every linked graduate with a PhD year of 2019 or later, we extract all positions held during calendar 2026, defined as positions that began on or before December 31, 2026 and either ended on or after January 1, 2026 or have no recorded end date.[5] We treat graduates who are linked to a profile but have no reported 2026 position as having uncertain employment, rather than as unemployed, and retain these graduates so that selection into the outcome sample can be examined directly. The extract contains 191,639 positions in 2026 for 145,043 graduates, and 14,254 linked graduates with no 2026 position. Our Revelio data run through June 23, 2026, such that "2026 employment" refers to positions held in the first half of the year.

We next classify each position in this sample along two dimensions: (i) the sector of the employer, and (ii) for academic employers, the type/track of the position.

[5]On LinkedIn, open-ended positions are common, and 95.7% of the positions in our extract have no end date.

Employer sector. Positions are grouped by employer, using Revelio's company identifier where present (94.6% of positions) and a normalized employer name otherwise, yielding 44,971 distinct employers. Each employer is assigned to one of eleven sectors: university; hospital or academic medical center; federally funded research and development center (FFRDC) or national laboratory; independent research institute; industry; government; non-profit; professional society; K–12 education; self-employed, independent, or founder; and unclassifiable. Classification is performed by a large language model ensemble in which two models independently classify every employer and a third adjudicates the cases on which they disagree, given the employer's name, web domain, industry (NAICS) code, parent company, and self-description as evidence.[6] The two primary models agree on 93.6% of employers, leaving 2,867 to adjudication. Against a blind hand-labeled sample of 50 employers, the ensemble's final label matches in 49 cases.

Academic position type. For positions held by PhDs at universities, hospitals (typically academic medical centers), and FFRDCs (namely, national labs)—which we collectively consider quasi-academic employment—we classify the raw job title into eight roles: tenure-track faculty, non-tenure-track faculty, postdoctoral researcher, research staff, clinical, administrative, graduate student or doctoral trainee, and other. We classify titles using the same LLM ensemble, providing it distinct (title, sector) pairs (23,453 pairs), with instructions to pay particular attention to the modifiers that indicate a tenure-track split (e.g., "Assistant Professor" is tenure-track, whereas "Research Assistant Professor," "Teaching Assistant Professor," "Clinical Assistant Professor," and "Assistant Professor of the Practice" are not). We evaluate this classification in a manually-labeled sample of 50 titles, which agrees with the LLM classification in 44 cases, and where the six discrepancies involve administrative roles that do not enter the outcomes we study.

Stale doctoral positions. Graduates commonly list their doctoral program itself as a position ("Graduate Research Assistant," "PhD Candidate"), typically with no end date, such that it can appear as ongoing university employment in 2026. We remove such entries with two rules. First, any position whose title is classified as graduate student or doctoral trainee is dropped, at any employer (13,888 positions). Second, any position at the graduate's own PhD institution that began before the year of the degree is dropped (17,020 positions). Together these rules remove 18,789 positions, leaving 135,911 graduates with at least one surviving 2026 position.

Outcome variables. From the surviving positions, we construct, for each graduate, indicators for holding any 2026 position at an employer in each sector, with "academic" employment defined as any of university, hospital or academic medical center, or FFRDC employment. The resulting sector indicators are not mutually exclusive: graduates who changed employers in 2026, or with concurrent appointments, can be in more than one sector, such that the indicators can add to more than one. Conditional on academic employment, we also create dummies for holding a tenure-track faculty, non-tenure-track faculty, or postdoctoral position. For graduates employed at a U.S. university, we additionally record whether the employer is an R1 institution and the employer's U.S. News

[6]The primary classifiers are `Claude Sonnet 5` (Anthropic) and `GPT-5.6 Terra` (OpenAI), with `Gemini 3.5 Flash` (Google) adjudicating disagreements. Prompts available on request.

& World Report and NRC program ranks, and where a graduate holds more than one university position, we take the best value across them—though these placement-quality measures are not currently being used in the analyses reported in this paper.

A.2.3 Coverage and selection

Coverage of LinkedIn is inevitably not random. If students who write with AI differ in their propensity to maintain a profile, the estimation sample for analyses of professional outcomes could potentially be selected on characteristics correlated with the treatment.

To assess this possibility, we examine selection at each step of our measurement funnel: whether a graduate is linked to a Revelio profile at all, whether the linked profile reports any 2026 position (conditional on having a profile), and whether any position is retained after filters to exclude reported-ongoing but likely-obsolete positions, particularly those at the graduate's PhD institution which appear vestigial to graduate school (conditional on having a position). Table A.3 reports regressions of each indicator on the AI-abstract indicator, with university-field and year fixed effects and standard errors clustered by university-field, over the 2023-2025 cohorts used in the paper's employment analysis. Graduates with AI-written abstracts are about three percentage points more likely to be linked to a profile (on a base of 70%). Conditional on having a linked profile, however, they are no more or less likely to have a reported 2026 position, and conditional on a position only marginally (and statistically weakly) more likely to have a position that survives our cleaning. The main difference is thus that graduates with AI-written abstracts are more likely to be on LinkedIn—a pattern which is consistent with their increased tendency to enter industry. Selection of this size cannot account for the employment differences in Figure 3: trimming the AI-abstract group by the excess link rate (per Lee 2009) bounds the industry difference between 6 and 11 p.p. and the academic difference between -10 and -6 p.p., both of which are far from zero.

Two further limitations should be noted. First, a position's employer is self-reported and updated at the profile owner's discretion—such that a graduate who has changed jobs but not updated their profile will be recorded at the earlier employer. This is the main reason we exclude 2026 graduates from the employment analysis: restricting this analysis to 2023-2025 cohorts permits at least half a year for profile updates, which limits but does not eliminate this problem. Second, because the analysis is restricted to R1 graduates, among whom research careers are more common and AI use is lower than in the sample as a whole, the estimated employment differences may not generalize to graduates of other institutions. That said, over 80% of graduates in our data are from R1 universities, and we consider them our primary population of interest.

Table A.3: Selection into the employment sample

	(1) Linked to Revelio	(2) Reports a 2026 position	(3) Has a counted outcome
Has AI in abstract	0.030***	0.007	0.010*
	(0.006)	(0.005)	(0.005)
N	90297	62674	55859
R^2	0.12	0.06	0.08
Y mean	0.696	0.892	0.914

Notes: Table estimates differences in Revelio coverage among 2023-2025 graduates by whether the graduate has an AI-generated dissertation abstract. Column (1) estimates differences in whether the graduate is linked to a Revelio (LinkedIn) profile; Column (2), whether the linked profile reports at least one position during 2026 (conditional on a linked profile); and Column (3), whether at least one 2026 position survives the exclusion of stale doctoral-program entries and contributes to our measurement of employment outcomes (conditional on having a 2026 position). All specifications include university-field and year fixed effects. *, **, *** represent significance at the 0.1, 0.05, and 0.01 levels, respectively. Standard errors clustered by university-field in parentheses.

A.3 Measuring university AI policies

To evaluate how universities govern the use of AI (particularly generative AI) in dissertation research, we assemble a new dataset of AI policies from 146 R1 universities (using the 2021 Carnegie Classification), which together account for over 80% of the graduates in our sample. For each university, we collect central AI policies publicly posted as of October 16, 2026, focusing on rules issued by the university or its graduate school that apply to doctoral students. Rules issued by individual departments or programs, or applying only to undergraduates or to faculty and staff, are not used in this analysis. This data collection effort only captures publicly available documents and thus excludes those which require an institutional login to access—though by our assessment, most university policy handbooks appear to be publicly accessible.

Collecting policy documents. Between October 6 and October 16, 2026, we searched each university's website for AI-related provisions in its course catalogs and bulletins, graduate school regulations, academic integrity codes, thesis and dissertation guidelines, research office guidance, and central AI, IT, and data security policies, archiving each relevant document together with its extracted text. Our catchment is intentionally broad, as generative AI use may be governed by policies across a wide range of functions. In addition to navigating each university's website, we also ran a fixed protocol of ten search queries per university (Table A.4) and screened the first 20 results of each query to identify additional relevant documents. We also used LLM-assisted web searches to identify additional potentially relevant policy documents, which we independently verified against the original sources before including them in our analysis. This process yielded 14,894 documents, of which 857 contain information useful for our classification.

Table A.4: Search queries used to collect university AI policies

#	Query
1	`"{University}" PhD dissertation AI policy`
2	`site:{domain} dissertation "artificial intelligence"`
3	`site:{domain} thesis "generative AI" guidelines`
4	`site:{domain} "graduate school" AI writing`
5	`site:{domain} AI (disclosure OR acknowledgment OR acknowledgement OR declaration)`
6	`site:{domain} "academic integrity" ("artificial intelligence" OR ChatGPT)`
7	`site:{domain} research "generative AI" (writing OR authorship)`
8	`site:{domain} filetype:pdf (handbook OR manual OR regulations) ("artificial intelligence" OR ChatGPT)`
9	`site:{domain} ("graduate council" OR "faculty senate") ("artificial intelligence" OR "generative AI")`
10	`"{University}" "AI policy" (graduate OR dissertation OR research) -site:{domain}`

Notes: Table lists the ten search queries run for each of the 146 universities between October 13 and 16, 2026, where `{University}` is the university's name and `{domain}` its web domain. Queries 1 and 10 search the web at large; queries 2 to 9 search the university's own domain for dissertation rules (2), thesis guidelines (3), graduate-school rules on AI writing (4), disclosure and acknowledgement requirements (5), academic-integrity provisions (6), research guidance on writing and authorship (7), handbooks, manuals, and regulations (8), and graduate council or faculty senate actions (9). For each query, we screened the first 20 results (or all results if fewer than 20 were returned), opened relevant or potentially relevant pages, examined their attachments, and searched for additional documents referenced within them. We supplemented these searches by directly navigating each university's website and using LLM-assisted web searches to identify additional potentially relevant policy documents. All documents identified through LLM-assisted searches were verified against their original sources before inclusion.

Classifying policies. We classify each university's policies on two dimensions: (a) whether the university has an explicit policy on AI use, and, if so, (b) what the policy requires.[7] With respect to the latter, we focus on measuring four non-mutually exclusive potential requirements: (i) obtaining approval for AI use (e.g., from a supervisor), (ii) disclosing of AI use, (iii) limiting permitted AI use to specified purposes, and (iv) complying with additional AI rules established by individual departments or programs. We classify a provision as a rule only if it explicitly addresses AI use and contains mandatory language (e.g., "must," "required," or "do not"), rather than advisory language (e.g., "should" or "encouraged"). General academic integrity policies or citation requirements that do not explicitly address AI are binned in a distinct category.

To assess the applicability of these rules to doctoral dissertations, we distinguish rules created specifically for dissertations from more general provisions. Where approval or disclosure requirements appear only in general provisions, we record the broadest scope of work covered: "research or graduate work" (including research, publications, or graduate-school rules extending beyond coursework), "general academic work" (such as institution-wide honor codes), or "coursework only" (including assignments, examinations, syllabi, or instructor permission).

Coding rules. We code each university separately for each requirement based on the contents of these documents, where every affirmative case must be supported by a quotation from a retrieved policy document. For each of these university requirements we code the level of specificity at

[7] We perform policy classification with the assistance of three state-of-the-art LLM agents from different frontier providers — `Claude Opus 5.5, Gemini Agron 4, and GPT-6 Astra`. Each agent codes every university from the retrieved documents independently, and we resolve disagreements among their codings over several review rounds by checking the cited quotations against the source documents rather than by majority vote.

which it applies. In particular, we count a provision as dissertation-specific only if it is specifically written for theses or dissertations and states what is required or who decides; rules that merely list theses among the work they cover, or mention only sanctions, are treated as general. We code requirements at the most specific level at which they are imposed, and universities at the level of their most specific AI rules. Duke, for example, is coded as having a dissertation-specific rule that requires both approval and disclosure, because its graduate school requires that any use of generative AI in a thesis or dissertation be discussed in advance with the advisor and committee and disclosed in the dissertation itself, even though it also has general course and data rules.[8] The University of Memphis, by contrast, has no dissertation rule; its university AI policy requires the course instructor's permission to use generative AI on exams and assignments, and requires that notable uses be acknowledged when an assignment is submitted, such that the university is coded as requiring approval and disclosure of AI use in coursework only.[9]

Although universities formulate their AI rules in different ways, we attempt to apply the same conventions to every university. For example, we count a rule as an approval requirement if AI use requires the permission of, or prior consultation with, an instructor, thesis committee, or adviser, including bans on disallowed uses. We count a rule as a disclosure requirement if it requires students to acknowledge, cite, or declare their AI use, or requires programs that allow AI to specify how such use must be disclosed, whereas rules that require disclosure only when an instructor, journal, or another policy asks for it are recorded separately. We count a rule as a restriction if it limits what AI may be used for (e.g., editing only) or requires that the work remain the student's own; general principles (e.g., that AI "must not replace critical thinking") and clauses that only require permission do not count, and bans on entering confidential data into AI tools are recorded separately as data restrictions. We count a rule as a program rule only if the university explicitly allows departments or programs to set their own AI rules, or requires students to follow them. Finally, policy drafts posted by a university are treated as current policy.

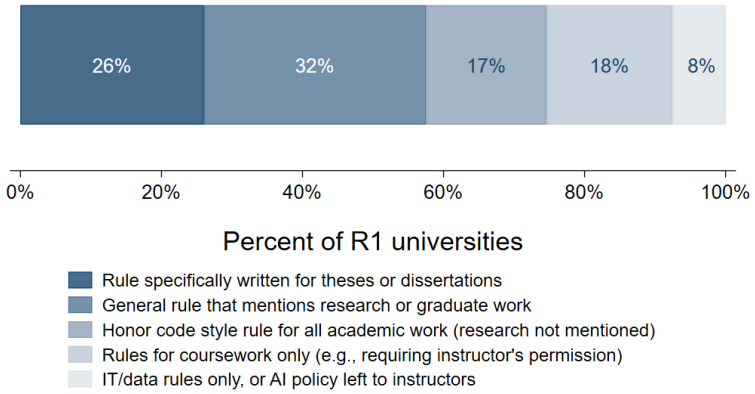
Resulting measures. Every R1 university in our sample has at least one central rule governing AI use in some form or capacity. Classifying each university by its most specific rule, 38 (26%) have a rule written specifically for theses or dissertations, 46 (32%) a general rule that mentions research or graduate work, 25 (17%) an honor-code style rule covering all academic work without mentioning research, 26 (18%) rules for coursework only, and 11 (8%) only IT or data-security rules or policies that leave requirements around AI use to instructors' discretion (Figure A.4). Requirements are

[8] Duke University Graduate School, *Guide for Electronic Submission of Theses and Dissertations* (last updated October 2025), p. 12: "Any use of generative Artificial Intelligence (AI) in producing the scholarly content of the thesis or dissertation must be discussed in advance among the student, advisor and committee, and must be fully disclosed and explained in a preface to the relevant chapter(s)." Available at https://gradschool.duke.edu/sites/default/files/2022-03/ETD%20Guide%202020.pdf (accessed October 6, 2026).

[9] University of Memphis, Policy AA3037, *Artificial Intelligence Tools* (last revised July 2024), p. 1: "Students may not use generative AI tools to complete exams or assignments (including papers, projects, computer programs, oral presentations, or other course requirements), unless given explicit permission from their course instructor." p. 2: "Regardless of the course AI guidelines, all notable uses of generative AI must be acknowledged in writing or via citation at the time of assignment submission." Available at https://www.memphis.edu/_resources/onspring/policy_pdf.php?asset=pdf&id=32 (accessed October 11, 2026).

common but mostly generic (Figure 4, Panel B). Disclosure is required at 117 universities (80%) and approval at 122 (84%), but only 21 and 25 of these, respectively, impose the requirement in a dissertation-specific provision. A further 71 require disclosure and 28 require approval in general rules covering research or all academic work, while the remainder apply only to coursework (19 for disclosure and 69 for approval, in which case the course instructor decides) or, for six universities, require disclosure only when an instructor, journal, or other policy does. Restrictions on specific uses appear at 130 universities (89%), but for 108 of them the only restriction is a ban on entering confidential data into unapproved AI tools; 22 limit what AI may do in academic work (for example, permitting editing only, or requiring that the work remain the student's own). 38 universities (26%) allow departments or programs to set their own AI rules.

Figure A.4: Distribution of R1 institutions by most specific dissertation AI policy



Notes: Figure shows the share of R1 institutions in our sample with AI policies of various degrees of specificity to dissertation research, as of October 2026.

Two measurement limitations should be noted. First, the absence of evidence is not evidence of absence: we may miss rules posted behind university logins or issued by units we do not treat as central policymaking bodies, such that the counts should be read as lower bounds on the prevalence of each policy and requirement—though as previously noted, most university-level policy documents appear to be public, and our sense is thus that our measurement is relatively complete. Second, the data are a snapshot of policies as of October 2026 and do not indicate when they were adopted, and we measure rules as they are written rather than as enforced.

A.4 Measuring dissertation AI acknowledgments

Our main measure of AI use, based on the detection of AI-generated text in dissertation abstracts, captures how students write. In light of the evidence of disclosure requirements, we supplement this evidence with information on disclosure patterns. To measure what students disclose, we identify and evaluate statements in full text dissertations where the author acknowledges using generative AI (Figure 4C). We analyze dissertation text using ProQuest's TDM Studio platform, through which we can access structured XML files with the contents of individual dissertations, including full text. Within the PQDT subset of our data (208,600 dissertations, or about 94% of our sample, which

also includes dissertations from university repositories), we have full text for 191,500 (92%), where coverage is 95-98% through 2023 but declines to 88% in 2024, 77% in 2025, and 69% in 2026, likely reflecting embargoes on the publication of recent dissertations.

Identifying candidate statements. After retrieving full text of each dissertation, we remove the ProQuest cover page, split the remaining text into sentences, and retain every sentence that names a generative AI model or tool, identified through keyword searches.[10] Names that also denote people or other objects (e.g., Claude, Gemini, Copilot) are retained only when the sentence or a neighboring sentence refers to AI. Most sentences mentioning AI tools describe AI as a subject of research rather than its use by the dissertation author. To identify potential disclosures of the author's own AI use, we next apply a rule-based filter that retains extracted sentences meeting at least one of three criteria: (i) containing first-person descriptions of AI use or explicit disclosure language, (ii) appearing in front matter (acknowledgments, prefaces, declarations, or AI-use statements), or (iii) mentioning an AI tool in connection with a writing task. After capping each dissertation at 60 such sentences, prioritizing those in front matter or containing disclosure language, our filter retains ≈14,000 candidate sentences from ≈3,500 dissertations. We export each candidate sentence together with its preceding and following sentences for context.

Classification. We classify each candidate sentence using a large language model, which receives the sentence, its neighboring sentences, its section heading, and the matched keywords, and assigns one of six labels: use of generative AI in producing the dissertation (e.g., drafting, editing, translation, literature search, coding, or figures); use of AI as an object or instrument of the research; a declaration that no AI was used; citation of an AI tool as a source; other mention; or a non-AI meaning of the keyword.[11] We code a dissertation as acknowledging AI use if at least one of its sentences is classified as use in producing the dissertation, except where the only tool named is Grammarly, which predates generative AI. This yields ≈700 acknowledging dissertations. As a placebo test, we apply the same procedure to dissertations from 2018-2022, before the widespread adoption of generative AI tools. Among the ≈113,000 full texts from this period, we identify only one acknowledgment: a 2021 dissertation thanking GPT-3 for providing writing prompts.

Table A.5 provides examples of AI disclosures in a subsample of recent dissertations, including an affirmative disclosure of no AI use. Table A.6 reports the annual share of dissertations with disclosed AI use. Acknowledgments of AI use rise from 0.20% of 2023 dissertations to 0.5% in 2024, 1.4% in 2025, and 4.7% in 2026. Most disclosed uses relate to editing and grammar, followed by coding and drafting text, and most disclosures name ChatGPT.

[10] The keywords for identifying potential disclosure statements are ChatGPT, GPT (with version suffixes), Claude, DeepSeek, Qwen, OpenAI, Anthropic, Copilot, Gemini, Bard, Llama, Mistral, Perplexity, Grammarly, QuillBot, Elicit, "large language model" (LLM), "generative AI," and phrases such as "AI-assisted" and "AI tool." The bare terms "AI" and "artificial intelligence" are not used as keywords, as they appear in a large share of science and engineering dissertations for reasons unrelated to disclosure.

[11] We use `GPT-4o` (OpenAI) through its Batch API, at temperature 0 and with structured output. The prompt defines each label and instructs the model to judge only from the text shown, and to prefer the first label when a sentence discloses both use in producing the dissertation and use in the research.

This measure should be interpreted as a lower bound on disclosed AI use, as it misses statements that name no AI tool and avoid the generic phrases in our keyword list, as well as statements filed outside the deposited document. The 2026 share is also concentrated. At three large institutions, 80-100% of 2026 dissertations include an AI-use statement (whether acknowledging use or declaring non-use), and manual review of dissertations from these schools reveals that each has a required template for reporting AI use. Together, these three institutions account for one-third of the identified AI acknowledgments in 2026, and without them, the 2026 acknowledging share is 3.2%—one-tenth of the share of dissertations we detect as having AI writing.

Table A.5: Example generative AI disclosures in 2026 dissertations

Dissertation ID	Statement
Example statements of use:	
3307301314	I would like to acknowledge the use of OpenAI's ChatGPT (GPT-5 model) for assistance with phrasing and editing portions of the literature review, particularly in enhancing clarity and readability. All intellectual contributions, analyses, and interpretations presented in this work are entirely my own and were verified against primary journal articles and scholarly sources.
3331860396	The generative AI model "Gemini", published by Google, was utilized to aid in ... the following ways. 1. To format bibtex citations of referenced works and for general LATEX, Overleaf, and Vim troubleshooting. 2. To act as a step in the process of verifying originality of potentially new results. 3. To aid in typesetting of names involving characters not in the standard 26 letter alphabet.
3343616927	Claude AI was used for paraphrasing in Chapter 1 and Chapter 3.
3344335001	During the preparation of this work the author(s) used ChatGPT-4o in order to assist with language refinement, improving clarity and expressions. ChatGPT-4o was not used to generate research data, analysis, or core content.
3345957149	ChatGPT was used as an assistive technology during the preparation and revision of this dissertation. The assistance was limited to language editing, improving clarity and organization, formatting support, LaTeX troubleshooting, and technical writing support. It was also used to discuss and refine the presentation of selected experimental results. The tool did not replace the author's scholarly judgment, experimental work, validation of results, or responsibility for the final dissertation content.
Affirmative statements of non-use:	
3292267286	I wish to state unequivocally that the technology currently known as generative artificial intelligence was not utilized in any capacity in the preparation of this dissertation, nor in any of the work described therein. What you are reading is purely the product of human effort, intellect, knowledge, and creativity.

Table A.6: PQDT dissertations acknowledging generative AI use, by year

Year	In PQDT sample	With full text	Disclosing AI use	Share disclosing (%)
2019	29,729	29,005	0	0.0
2020	28,540	27,727	0	0.0
2021	28,936	27,900	1	0.0
2022	29,901	28,426	0	0.0
2023	29,748	28,401	54	0.2
2024	28,914	25,358	138	0.5
2025	27,058	20,731	295	1.4
2026	5,785	3,964	186	4.7

Notes: Table reports the number and share of dissertations in our PQDT sample in which the author acknowledges using generative AI to prepare the dissertation, by year. The sample consists of U.S. PhD dissertations in the natural sciences and engineering published between 2019 and 2026 in PQDT. Shares are computed over dissertations whose full text is available in ProQuest TDM Studio. Statements naming only Grammarly are not counted as acknowledgments. Because we sampled PQDT in mid-2026, data for the 2026 cohort are incomplete, and because some dissertation texts only become available after an initial embargo period, full text availability rates are lower in the most recent years.

B Supplementary Results

B.1 AI use by university and program

Tables B.1 and B.2 list the universities and university-fields with the fewest and most students with AI writing. Both tables are restricted to R1 universities and 2025-2026 graduates, and report the number of such graduates and share with AI-generated abstracts. Table B.1 lists the ten universities with the lowest and highest shares among those with at least 50 graduates over this period; Table B.2 does the same for university-fields with at least 20 graduates. The lowest rates are concentrated at a small number of highly selective institutions and in mathematics, physics, and chemistry, whereas the highest are spread across a wider set of universities and fields, with computer science and engineering appearing most often.

Table B.1: Top/bottom universities by share of AI dissertations, 2025-2026 (R1 only)

Lowest share			Highest share		
University	N	Percent AI	University	N	Percent AI
U. of Massachusetts-Amherst	51	4%	U. of Idaho	90	28%
U. of Oregon	123	9%	Lehigh U.	131	28%
Massachusetts Inst. of Tech.	389	9%	Arizona State U.	529	29%
Syracuse U.	84	10%	Saint Louis U.	57	30%
Boston Coll.	61	10%	U. of Nevada-Reno	117	30%
California Inst. of Tech.	151	10%	U. of Southern California	73	30%
Ohio State U.	285	12%	U. of Memphis	82	30%
Georgetown U.	67	12%	U. of North Texas	81	33%
U. of Houston	66	12%	U. of Wyoming	75	35%
Princeton U.	444	13%	Virginia Commonwealth U.	50	36%

Notes: Table lists universities with the lowest and highest shares of their post-2025 dissertations with AI abstracts. Analysis restricted to R1 universities with at least 50 graduates over this period. Table reports, for each university, the number of graduates and the AI share.

Table B.2: Top/bottom university-fields by share of AI dissertations, 2025-2026 (R1 only)

Lowest share				Highest share			
University	Field	N	Percent AI	University	Field	N	Percent AI
U. of California-Berkeley	Maths	33	0%	U. of Florida	Agri. Sci.	21	48%
Ohio State U.	Chemistry	20	0%	Temple U.	Comp. Sci.	21	48%
Rice U.	Maths	20	0%	Rutgers U.	Comp. Sci.	48	50%
Northwestern U.	Bio. Eng.	29	0%	U. of Illinois Chicago	Oth. Eng.	20	50%
Massachusetts Inst. of Tech.	Physics	57	2%	U. of California-Davis	Comp. Sci.	21	52%
U. of Minnesota-Twin Cities	Health Sci.	34	3%	Northeastern U.	Health Sci.	66	53%
U. of California-Santa Barbara	Physics	31	3%	Indiana U.-Bloomington	Comp. Sci.	28	54%
Virginia Tech	Bio. Sci.	32	3%	Iowa State U.	Comp. Sci.	20	55%
Massachusetts Inst. of Tech.	Health Sci.	29	3%	U. of California-Santa Cruz	Comp. Sci.	26	58%
U. of Colorado Boulder	Maths	28	4%	U. of California-Irvine	Oth. Eng.	26	58%

Notes: Table lists university-fields with the lowest and highest shares of their post-2025 dissertations with AI abstracts. Analysis restricted to R1 universities and to university-fields with at least 20 graduates over this period. Table reports, for each university-field, the number of graduates and the AI share.

B.2 Peer AI use as a predictor of individual use

The patterns in Figure 2 raise the question of whether AI use is a property of individual students or of the environments in which they train. To evaluate this question, we relate a graduate's own AI use to that of their immediate predecessors in the same program or under the same primary advisor (committee chair). We estimate the following specification:

$$AI_{it} = \beta \cdot PeerAI_{p,t-1} + \delta_t + \varepsilon_{it}. \tag{B.1}$$

where i indexes graduates, p indexes peer groups (a doctoral program, defined as a university-field, or an advisor), and t indexes graduation years; AI_{it} is an indicator for an AI-generated abstract; $PeerAI_{p,t-1}$ is the average AI share of abstracts among the previous year's graduates of the same peer group; and δ_t are year fixed effects, with an estimation sample of 2023 to 2025 graduates with at least one same-group graduate in the preceding year.

Table B.3 reports estimates with the program-based measure (Columns 1 to 3) and the advisor-based measure (Columns 4 to 6), in each case sequentially narrowing the sample from all institutions to R1 and R2 institutions to R1 institutions only. The estimates are stable across samples: a 10 p.p. higher AI share among the previous year's graduates of the same program is associated with a roughly 3 p.p. higher probability that a graduate's own abstract is AI-generated, and the corresponding figure for the same advisor is roughly 2 p.p. We interpret these results as indicating that AI use clusters within training environments—whether due to peer influence, shared norms, common access, or advisor practices—though not as a source of exogenous variation in AI use per se, since these same environments could affect graduates' outcomes directly.

Table B.3: Same-program and same-advisor peer AI use predicts own AI use

	Program-based			Advisor-based		
	(1)	(2)	(3)	(4)	(5)	(6)
	All	R1/R2	R1 only	All	R1/R2	R1 only
Program AI share	0.305***	0.274***	0.277***			
	(0.036)	(0.038)	(0.044)			
Advisor AI share				0.183***	0.176***	0.172***
				(0.024)	(0.026)	(0.029)
N	88838	82141	73705	33667	31195	28400
R^2	0.07	0.07	0.07	0.08	0.08	0.08
Year FEs	Y	Y	Y	Y	Y	Y

Notes: Table shows that an individual's use of AI can be predicted from (i) the rate at which prior-year students in the same doctoral program (measured by university-field) use AI (Columns 1 to 3) and (ii) the rate at which prior-year students with the same advisor use AI (Columns 4 to 6). Within each prediction set, we sequentially narrow the sample from all institutions, to R1 and R2 institutions, to R1 institutions only; patterns are similar across them. Sample restricted to 2023-2025 graduates with at least one prior-cohort peer in the same program or with the same advisor. *, **, *** represent significance at the 0.1, 0.05, and 0.01 levels, respectively. Standard errors clustered by doctoral program in parentheses.

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